

AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

47

Fuel Tank (large fabricated shell)

1528 × 658 × 594 mm · 10595.9 cm³ · AI 28.609 kg / Steel 83.178 kg

Manufacturability: 47/100 · Confidence: Low

§1 — Geometry & Part Summary

Bounding Box	1528.3 × 657.6 × 593.7 mm	Surface Area	47571.3 cm ²
Volume	10595.90 cm ³	Weight (AI)	28.609 kg
Weight (Steel)	83.178 kg	Weight (Plastic)	11.126 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Bend/Form lines	372	High	Multiple bend transitions consistent with a deep, multi-panel formed tank shell
Pierced/threaded holes	8	Medium	Fitting/port holes, some threaded for fuel line/sender connections
Bosses/necks (r>=30mm)	12	Medium	Filler neck, sender flange or drain boss features
Free-form/B-spline surfaces	694	High	Organic tank contours (694 B-spline faces) typical of blow/roto-moulded or deep-drawn double-curvature shell
Undercuts	753	High	753 undercut faces — indicates re-entrant flange/weld-lip geometry not achievable in a single-direction stamping/forming operation

§3 — Material Analysis

AI-suggested

Primary Material:

DC01 Cold-Rolled Mild Steel Sheet (55% confidence)

No explicit aluminium signal in filename/geometry. Fuel tanks of this size (1.5m) and production volume (200k/yr) are most commonly mild/terne-coated steel for cost and weldability. Wall thickness range (4.15-18.34mm measured, incl. local bosses/flanges) is consistent with formed-welded steel gauge (~3-5mm) plus reinforced boss inserts. Per guidance, defaulting to steel over aluminium absent an explicit lightweighting requirement.

Alternatives:

Aluminium 5052 Sheet (30%) · Stainless Steel 316 (10%)

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Progressive die stamping + multi-panel forming	sheet_metal	55%	0.0017	Stage-1 classifier and deterministic tooling inputs (blank/strip/pitch) point to progressive-die sheet metal forming; however the part size (1.5m) exceeds typical single progressive-die envelopes and likely requires multi-panel forming + weld assembly.
Sheet metal fabrication (press-brake/roll-form + TIG/MIG weld assembly)	sheet_metal_fab	40%	0.0500	Given the large envelope and complex double-curvature shell, a welded multi-panel fabrication (formed shells welded at seams) is a realistic alternative to a monolithic progressive die.

§5 — Manufacturability Risks (Score: 47/100)

Severity	Feature / Area	Description	Recommended Action
High	Undercuts (753 faces)	Large number of undercut/re-entrant features incompatible with straight-draw stamping or single-axis forming; will require secondary forming, side-actions, or welded sub-assemblies.	Split into multiple simpler formed shells joined by seam welding; avoid re-entrant flanges in a single draw.
High	372 bend operations	Extremely high bend count for one panel increases springback variability, die wear, and station count/cost in a progressive die.	Consolidate bends, redesign ribs/flanges to reduce discrete bend count, or move to roll-forming for long straight bends.
Medium	Non-uniform wall (4.15-18.34mm)	Wide thickness variation is atypical for pure sheet metal — indicates thick bosses/flanges likely requiring welded-in machined inserts.	Use separate machined/cast boss inserts welded onto the thin-gauge shell rather than attempting to form thick sections from sheet.
Medium	Large blank size (1605x691mm)	Requires a very large press bed and correspondingly large/expensive progressive die tooling; press tonnage and handling become significant cost drivers.	Confirm press capacity early; consider tandem-line stamping of smaller panels with post-weld assembly.
Low	Free-form double-curvature (694 B-spline faces)	Complex 3D curvature increases die machining cost and risk of wrinkling/tearing during deep forming.	Simplify curvature where possible; use draw beads and controlled blank-holder force in die design.

§6 — DFM Issues (sheet_metal)

Severity	Area	Description	Impact	Fix
High	Undercut/re-entrant flanges	753 undercut faces cannot be produced by a single-direction stamping or forming operation.	Forces multi-piece welded construction, adding weld stations, leak-test cost, and cycle time.	Redesign flanges/ports to be accessible from a single draw direction, or explicitly split into weld-assembled sub-shells.
High	Bend count (372)	Very high discrete bend count for a single blank drives die complexity, station count, and springback variation across a 1.6m part.	Increases progressive die cost, cycle time, and scrap/rework rate.	Combine adjacent bends into single-radius forms; move long straight bends to roll-forming.
Medium	Non-uniform wall thickness	Local sections up to 18.34mm thick are inconsistent with a formed sheet gauge (mean ~5mm), suggesting welded boss/flange inserts.	Adds secondary machining/welding operations and risk of weld-zone leaks on a pressure-containing tank.	Specify machined or cast boss inserts welded onto a uniform-gauge shell rather than forming thick sections directly from sheet.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£0.00	£0.00	£0.00

Net Weight	83.178 kg	Material	mat-dc01
Recommended Process	sheet_metal	Cycle Time	7.1390 hr/part
Setup Time	0.750 hr	Operations	2 ops
Operations Detail	Progressive die blank/form/pierce (mach-haas-vf2, 0.0017 hr) Secondary drill/tap for threaded fittings (mach-drill, 0.0670 hr)		

Process-Specific Parameters

Casting Subtype	sand
Die/Mould Cost	£0
Die Life	0 shots
Cavities	0
Yield	0.0%
Flash Weight	0.000 kg
Yield	0.0%
Die Cost	£0
Strokes	0
Cavities	0
Wall Thickness	0 mm
Mould Cost	£0
Mould Life	0 shots
Projected Area	0.0 cm²

§8 — AI Analysis Explanation